INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

Salt Lake Campus

Under University of Engineering & Management, Kolkata

Degree: BCA , Stream:BCA

Term –II, First Semester Examination,November-2023

Subject Code: BCAMD101

Subject Name: Digital Electronics

Full Marks:30 Duration:1 Hour

Part - A

Attempt 5 questions

Each question carries 2 Marks (2 X 5)

1.	What is meant by parallel shifting?	2
	or	
1.	State the difference between a register and a counter.	2
2.	What is the function of select inputs of a multiplexer?	2
	or	
2.	Perform the BCD subtraction: $(4532)_{10} - (2671)_{10}$	2
3.	Define Propagation Delay.	2
	or	
3.	What is Parallel Adder?	2
4.	When do we use (r-1)'s complement method to find the complement of a number?	2
	or	
4.	Illustrate 10's complement subtraction for 753-864	2
5.	Determine the number of flip-flops that would be required to build Mod-19 counters and draw the state diagram	2
	or	
5.	Demonstrate the primary disadvantage of an asynchronous counter.	2

Part - B
Attempt 2 questions
Each question carries 5 Marks (5 X 2)

6. Convert D flip-flop to T flip-flop. 5

or

6. Evaluate an asynchronous 3-bit up-down counter. 5

7. Evaluate 10's complement of 935 base 11. 5

or

7. Design a JK flip-flop from SR flip-flop 5

Part - C
Attempt 1 question
Each question carries 10 Marks (10 X 1)

8. Illustrate the working of Parallel In Serial Out shift register with neat logic diagram. 10

or

8. Analyse the following the condition and explain it with proper circuit: 10
   - 16:1 multiplexer using two 8:1 multiplexer
   - Multiplexer to implement the Boolean expression

$$F(A,B,C,D) = \sum(1,3,4,11,12,13,15)$$